
Corrigé — Exercice 10

1) $\det(A) = 0 \cdot (0 - 1) - 1 \cdot (0 - 1) + 1 \cdot (1 - 0) = 0 + 1 + 1 = 2 \neq 0$: A inversible.

2) $A^2 = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$; or $A + 2I_3 = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$, donc $A^2 = A + 2I_3$.

De $A(A - I_3) = 2I_3$ on tire $A^{-1} = \frac{1}{2}(A - I_3) = \frac{1}{2} \begin{pmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{pmatrix}$.

3) Le système s'écrit $AX = \begin{pmatrix} 3 \\ 5 \\ 4 \end{pmatrix}$, donc $X = \frac{1}{2}(A - I_3) \begin{pmatrix} 3 \\ 5 \\ 4 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 6 \\ 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$.

$$(x, y, z) = (3, 1, 2)$$